

Markowitz2EPG

$$L(\vec{\omega}) = \vec{\omega}^T \Sigma \vec{\omega} - \lambda \mu^T \vec{\omega} \quad (1)$$

- $\vec{\omega}$: Vektor bobot aset.
- Σ : Matriks kovariansi dengan elemen σ_{ij} .
- μ : Matriks *expected return* dengan elemen μ_i .
- λ : Parameter *risk tolerance*.

$$\lambda = \frac{2}{\gamma} \implies \frac{1}{\lambda} = \frac{\gamma}{2} \quad (2)$$

- γ : Parameter *risk aversion*.

$$\vec{\omega} = \frac{\vec{x}}{k}, \quad k \in \mathbb{R} \quad (3)$$

- $\vec{x}$: Vektor strategi dalam *game*.
- k : Parameter skalar (mengingat $\sum_i \omega_i = \sum_i \frac{x_i}{k}$).

$$L(\vec{x}) = \left(\frac{\vec{x}}{k} \right)^T \Sigma \left(\frac{\vec{x}}{k} \right) - \lambda \mu^T \left(\frac{\vec{x}}{k} \right) \quad (4)$$

$$L(\vec{x}) = \frac{1}{k^2} \vec{x}^T \Sigma \vec{x} - \frac{\lambda}{k} \mu^T \vec{x} \quad (5)$$

$$\vec{x}^T \Sigma \vec{x} = \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j \quad (6)$$

$$\mu^T \vec{x} = \sum_{i=1}^N \mu_i x_i \quad (7)$$

$$U(\vec{x}) = L(\vec{x}) \cdot \left(-\frac{1}{\lambda} \right) \quad (8)$$

$$U(\vec{x}) = \left(\frac{1}{k^2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j - \frac{\lambda}{k} \sum_{i=1}^N \mu_i x_i \right) \cdot \left(-\frac{\gamma}{2} \right) \quad (9)$$

$$U(\vec{x}) = \frac{1}{k} \sum_{i=1}^N \mu_i x_i - \frac{\gamma}{2k^2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j \quad (10)$$

- $U(\vec{x})$: Fungsi potensial *Exact Potential Game*.

$$u_i(x_i, \vec{x}_{-i}) = U(x_i, \vec{x}_{-i}) - U(0, \vec{x}_{-i}) \quad (11)$$

$$u_i(x_i, \vec{x}_{-i}) = x_i \left(\frac{\mu_i}{k} - \frac{\gamma \sigma_{ii} x_i}{2k^2} - \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \right) \quad (12)$$

Skenario A: $x_i = 0 \rightarrow x'_i = 1$

$$\Delta u_i(0 \rightarrow 1) = \frac{\mu_i}{k} - \frac{\gamma \sigma_{ii}}{2k^2} - \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \quad (13)$$

Skenario B: $x_i = 1 \rightarrow x'_i = 0$

$$\Delta u_i(1 \rightarrow 0) = -\frac{\mu_i}{k} + \frac{\gamma \sigma_{ii}}{2k^2} + \frac{\gamma}{k^2} \sum_{j \neq i} \sigma_{ij} x_j \quad (14)$$

- Δu_i : Perubahan utilitas marginal pemain i .

Kasus Khusus ($k = 1$):

$$\Delta u_i|_{k=1} = \mu_i - \frac{\gamma}{2} \sigma_{ii} - \gamma \sum_{j \neq i} \sigma_{ij} x_j \quad (15)$$

Transformasi QUBO

$$E(\vec{x}) = -U(\vec{x}) \quad (16)$$

$$E(\vec{x}) = - \left(\frac{1}{k} \sum_{i=1}^N \mu_i x_i - \frac{\gamma}{2k^2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j \right) \quad (17)$$

$$E(\vec{x}) = \frac{\gamma}{2k^2} \sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j - \frac{1}{k} \sum_{i=1}^N \mu_i x_i \quad (18)$$

$$\sum_{i=1}^N \sum_{j=1}^N \sigma_{ij} x_i x_j = \sum_{i=1}^N \sigma_{ii} x_i^2 + \sum_{i \neq j} \sigma_{ij} x_i x_j \quad (19)$$

$$x_i^2 = x_i, \quad x_i \in \{0, 1\} \quad (20)$$

$$E(\vec{x}) = \sum_{i=1}^N \left(\frac{\gamma \sigma_{ii}}{2k^2} - \frac{\mu_i}{k} \right) x_i + \frac{\gamma}{2k^2} \sum_{i \neq j} \sigma_{ij} x_i x_j \quad (21)$$

$$P(\vec{x}) = A \left(\sum_{i=1}^N x_i - k \right)^2 \quad (22)$$

$$P(\vec{x}) = A \left[\left(\sum_{i=1}^N x_i \right)^2 - 2k \sum_{i=1}^N x_i + k^2 \right] \quad (23)$$

$$\left(\sum_{i=1}^N x_i \right)^2 = \sum_{i=1}^N x_i^2 + \sum_{i \neq j} x_i x_j = \sum_{i=1}^N x_i + \sum_{i \neq j} x_i x_j \quad (24)$$

$$P(\vec{x}) = \sum_{i=1}^N A(1 - 2k)x_i + \sum_{i \neq j} A x_i x_j + A k^2 \quad (25)$$

$$H(\vec{x}) = E(\vec{x}) + P(\vec{x}) \quad (26)$$

$$H(\vec{x}) = \sum_{i=1}^N \left[\frac{\gamma \sigma_{ii}}{2k^2} - \frac{\mu_i}{k} + A(1 - 2k) \right] x_i + \sum_{i \neq j} \left(\frac{\gamma \sigma_{ij}}{2k^2} + A \right) x_i x_j + A k^2 \quad (27)$$

$$H(\vec{x}) = \sum_{i=1}^N Q_{ii} x_i + \sum_{i \neq j} Q_{ij} x_i x_j + C \quad (28)$$

- $H(\vec{x})$: Hamiltonian total (QUBO).
- $Q_{ii} = \frac{\gamma \sigma_{ii}}{2k^2} - \frac{\mu_i}{k} + A(1 - 2k)$: Koefisien linear.
- $Q_{ij} = \frac{\gamma \sigma_{ij}}{2k^2} + A$: Koefisien kuadratik.
- $C = A k^2$: Konstanta energi.

Transformasi Ising

$$x_i = \frac{1 - s_i}{2}, \quad s_i \in \{-1, 1\} \quad (29)$$

$$H(\vec{s}) = \sum_{i=1}^N Q_{ii} \left(\frac{1 - s_i}{2} \right) + \sum_{i \neq j} Q_{ij} \left(\frac{1 - s_i}{2} \right) \left(\frac{1 - s_j}{2} \right) + C \quad (30)$$

dengan

$$\left(\frac{1 - s_i}{2} \right) \left(\frac{1 - s_j}{2} \right) = \frac{1}{4} (1 - s_i - s_j + s_i s_j) \quad (31)$$

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$$H(\vec{s}) = \sum_{i=1}^N \frac{Q_{ii}}{2} (1 - s_i) + \sum_{i \neq j} \frac{Q_{ij}}{4} (1 - s_i - s_j + s_i s_j) + C \quad (32)$$

$$H(\vec{s}) = \sum_{i \neq j} \frac{Q_{ij}}{4} s_i s_j - \sum_{i=1}^N \left(\frac{Q_{ii}}{2} + \sum_{j \neq i} \frac{Q_{ij}}{2} \right) s_i + \left(\sum_{i=1}^N \frac{Q_{ii}}{2} + \sum_{i \neq j} \frac{Q_{ij}}{4} + C \right) \quad (33)$$

$$H(\vec{s}) = \sum_{i \neq j} J_{ij} s_i s_j - \sum_{i=1}^N h_i s_i + C_{Ising} \quad (34)$$

- $J_{ij} = \frac{Q_{ij}}{4}$: Kekuatan interaksi antar *spin*.
- $h_i = \frac{Q_{ii}}{2} + \sum_{j \neq i} \frac{Q_{ij}}{2}$: Medan magnet eksternal.
- $C_{Ising} = \sum_{i=1}^N \frac{Q_{ii}}{2} + \sum_{i \neq j} \frac{Q_{ij}}{4} + C$: Pergeseran energi konstan.

$$s_i \rightarrow \hat{Z}_i \quad (35)$$

$$\hat{Z}_i = \mathbb{I}_2 \otimes \mathbb{I}_2 \otimes \cdots \otimes \sigma_i^z \otimes \cdots \otimes \mathbb{I}_2 \quad (36)$$

$$\sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \mathbb{I}_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (37)$$

$$\hat{H} = \sum_{i \neq j} J_{ij} \hat{Z}_i \hat{Z}_j + \sum_{i=1}^N h_i \hat{Z}_i + C_{Ising} \hat{\mathbb{I}} \quad (38)$$

$$E(\theta) = \langle \psi(\theta) | \hat{H} | \psi(\theta) \rangle \quad (39)$$

$$E(\theta) = \sum_{i \neq j} J_{ij} \langle \psi(\theta) | \hat{Z}_i \hat{Z}_j | \psi(\theta) \rangle + \sum_{i=1}^N h_i \langle \psi(\theta) | \hat{Z}_i | \psi(\theta) \rangle + C_{Ising} \quad (40)$$

Quantum Computing

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (41)$$

- $|0\rangle, |1\rangle$: Vektor basis komputasi kuantum (*computational basis states*).

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (42)$$

- $\sigma_x, \sigma_y, \sigma_z$: Matriks Pauli X, Y , dan Z .

$$\hat{R}_y(\theta) = \exp\left(-i\frac{\theta}{2}\sigma_y\right) \quad (43)$$

$$\hat{R}_y(\theta) = \hat{I} \cos\left(\frac{\theta}{2}\right) - i\sigma_y \sin\left(\frac{\theta}{2}\right) \quad (44)$$

$$\hat{R}_y(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) - i \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) \quad (45)$$

$$\hat{R}_y(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{pmatrix} \quad (46)$$

$$\hat{R}_z(\theta) = \exp\left(-i\frac{\theta}{2}\sigma_z\right) \quad (47)$$

$$\hat{R}_z(\theta) = \hat{I} \cos\left(\frac{\theta}{2}\right) - i\sigma_z \sin\left(\frac{\theta}{2}\right) \quad (48)$$

$$\hat{R}_z(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cos\left(\frac{\theta}{2}\right) - i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \sin\left(\frac{\theta}{2}\right) \quad (49)$$

$$\hat{R}_z(\theta) = \begin{pmatrix} \cos\left(\frac{\theta}{2}\right) - i \sin\left(\frac{\theta}{2}\right) & 0 \\ 0 & \cos\left(\frac{\theta}{2}\right) + i \sin\left(\frac{\theta}{2}\right) \end{pmatrix} \quad (50)$$

$$\hat{R}_z(\theta) = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} \quad (51)$$

$$\text{CNOT} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \sigma_x \quad (52)$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \otimes \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (53)$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (54)$$

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (55)$$

sehingga CNOT dengan kontrol q_0 dan target q_1 adalah

$$\text{CNOT}_{q_0, q_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (56)$$

$$\begin{aligned} |00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |01\rangle \\ |10\rangle &\rightarrow |11\rangle \\ |11\rangle &\rightarrow |10\rangle \end{aligned} \quad (57)$$

namun CNOT dengan kontrol q_1 dan target q_0 adalah

$$\text{CNOT}_{q_1, q_0} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (58)$$

$$\begin{aligned} |00\rangle &\rightarrow |00\rangle \\ |01\rangle &\rightarrow |11\rangle \\ |10\rangle &\rightarrow |10\rangle \\ |11\rangle &\rightarrow |01\rangle \end{aligned} \quad (59)$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (60)$$

- α, β : Amplitudo probabilitas ($\alpha, \beta \in \mathbb{C}$).

$$\langle\psi|\psi\rangle = 1 \quad (61)$$

$$(\alpha^*\langle 0| + \beta^*\langle 1|)(\alpha|0\rangle + \beta|1\rangle) = 1 \quad (62)$$

$$|\alpha|^2\langle 0|0\rangle + \alpha^*\beta\langle 0|1\rangle + \beta^*\alpha\langle 1|0\rangle + |\beta|^2\langle 1|1\rangle = 1 \quad (63)$$

$$|\alpha|^2 + |\beta|^2 = 1 \quad (64)$$

$$\hat{H}|\psi_n\rangle = E_n|\psi_n\rangle \quad (65)$$

- $\hat{H}$: Operator Hamiltonian.
- $|\psi_n\rangle$: Eigenstate ke- n .
- E_n : Eigenvalue energi ke- n .

$$E(\theta) = \frac{\langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle}{\langle\psi(\theta)|\psi(\theta)\rangle} \quad (66)$$

$$E(\theta) = \langle\psi(\theta)|\hat{H}|\psi(\theta)\rangle \quad (67)$$

- $E(\theta)$: Nilai ekspektasi energi (*cost function*).
- $|\psi(\theta)\rangle$: *Trial ansatz* terparameterisasi.

$$E(\theta) = \langle \psi(\theta) | \left(\sum_n E_n |\psi_n\rangle \langle \psi_n| \right) | \psi(\theta) \rangle \quad (68)$$

$$E(\theta) = \sum_n E_n |\langle \psi_n | \psi(\theta) \rangle|^2 \geq E_0 \quad (69)$$

- Prinsip Variasional Rayleigh-Ritz: $E(\theta) \geq E_0$.

Gradient Descent

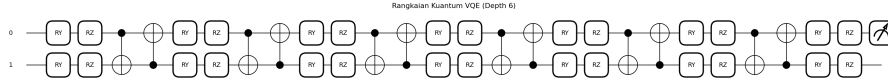


Figure 1: Representasi sirkuit kuantum variasional dengan struktur *ansatz* yang terdiri dari operator $\hat{U}_A$, $\hat{U}(\theta)$, dan $\hat{U}_B$ untuk optimasi parameter θ .

Untuk kasus 1 depth (8 parameter θ):

$$|\psi(\theta)\rangle = \hat{U}_{rot}^{(1)}(\theta_1) \cdot \hat{U} \cdot \hat{U}_{rot}^{(0)}(\theta_0) |00\rangle \quad (70)$$

dimana operator rotasi qubit j layer l didefinisikan sebagai $\hat{U}_{l,j} = R_z(\theta_{l,j,z})R_y(\theta_{l,j,y})$ dengan representasi matriks:

$$\hat{U}_{l,j} = \begin{pmatrix} e^{-i\frac{\theta_{l,j,z}}{2}} \cos \frac{\theta_{l,j,y}}{2} & -e^{-i\frac{\theta_{l,j,z}}{2}} \sin \frac{\theta_{l,j,y}}{2} \\ e^{i\frac{\theta_{l,j,z}}{2}} \sin \frac{\theta_{l,j,y}}{2} & e^{i\frac{\theta_{l,j,z}}{2}} \cos \frac{\theta_{l,j,y}}{2} \end{pmatrix} \quad (71)$$

dan $\hat{U}_{rot}^{(l)} = \hat{U}_{l,0} \otimes \hat{U}_{l,1}$. Operator entainment $\hat{U}_{ent}$ merupakan cascade CNOT:

$$\hat{U}_{ent} = CNOT_{q_1,q_0} \cdot CNOT_{q_0,q_1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{pmatrix} \quad (72)$$

Penurunan state dilakukan secara sekuensial. Aplikasi rotasi pertama pada $|00\rangle$ menghasilkan:

$$|\phi_0\rangle = \hat{U}_{rot}^{(0)} |00\rangle = \begin{pmatrix} e^{-i\frac{\theta_{0,0,z} + \theta_{0,1,z}}{2}} \cos \frac{\theta_{0,0,y}}{2} \cos \frac{\theta_{0,1,y}}{2} \\ e^{-i\frac{\theta_{0,0,z}}{2}} \cos \frac{\theta_{0,0,y}}{2} \sin \frac{\theta_{0,1,y}}{2} \\ e^{i\frac{\theta_{0,0,z} - \theta_{0,1,z}}{2}} \sin \frac{\theta_{0,0,y}}{2} \cos \frac{\theta_{0,1,y}}{2} \\ e^{i\frac{\theta_{0,0,z} + \theta_{0,1,z}}{2}} \sin \frac{\theta_{0,0,y}}{2} \sin \frac{\theta_{0,1,y}}{2} \end{pmatrix} \quad (73)$$

Aplikasi operator entrainment $\hat{U}_{ent}$ pada $|\phi_0\rangle$ menghasilkan $|\phi_1\rangle$:

$$|\phi_1\rangle = \hat{U}_{ent}|\phi_0\rangle = \begin{pmatrix} e^{-i\frac{\theta_{0,0,z}+\theta_{0,1,z}}{2}} \cos \frac{\theta_{0,0,y}}{2} \cos \frac{\theta_{0,1,y}}{2} \\ e^{i\frac{\theta_{0,0,z}-\theta_{0,1,z}}{2}} \sin \frac{\theta_{0,0,y}}{2} \cos \frac{\theta_{0,1,y}}{2} \\ e^{i\frac{\theta_{0,0,z}+\theta_{0,1,z}}{2}} \sin \frac{\theta_{0,0,y}}{2} \sin \frac{\theta_{0,1,y}}{2} \\ e^{-i\frac{\theta_{0,0,z}-\theta_{0,1,z}}{2}} \cos \frac{\theta_{0,0,y}}{2} \sin \frac{\theta_{0,1,y}}{2} \end{pmatrix} \quad (74)$$

State akhir $|\psi(\theta)\rangle$ diperoleh dengan mengaplikasikan rotasi layer kedua:

$$|\psi(\theta)\rangle = \hat{U}_{rot}^{(1)}|\phi_1\rangle = (\hat{U}_{1,0} \otimes \hat{U}_{1,1})|\phi_1\rangle \quad (75)$$

$$\hat{U}(\theta) = \mathbb{I} \otimes \hat{R}_y(\theta) = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} & 0 & 0 \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} & 0 & 0 \\ 0 & 0 & \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ 0 & 0 & \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \quad (76)$$

$$\hat{U}_B = \prod_{j=k+1}^m \hat{G}_j, \quad \hat{G}_j \in \mathbb{C}^{4 \times 4} \quad (77)$$

dimana $\hat{G}_j$ merupakan operator unitari (gate) pada sirkuit post-processing.

$$\hat{U}(\theta) = e^{-i\frac{\theta}{2}\hat{P}} \quad (78)$$

$$|\phi\rangle = \hat{U}_A|0\rangle \quad (79)$$

$$\hat{M} = \hat{U}_B^\dagger \hat{H} \hat{U}_B \quad (80)$$

$$E(\theta) = \langle \phi | \hat{U}^\dagger(\theta) \hat{M} \hat{U}(\theta) | \phi \rangle \quad (81)$$

$$\frac{\partial \hat{U}(\theta)}{\partial \theta} = -i\frac{1}{2}\hat{P}\hat{U}(\theta) \quad (82)$$

$$\frac{\partial \hat{U}^\dagger(\theta)}{\partial \theta} = i\frac{1}{2}\hat{U}^\dagger(\theta)\hat{P} \quad (83)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \langle \phi | \frac{\partial \hat{U}^\dagger(\theta)}{\partial \theta} \hat{M} \hat{U}(\theta) | \phi \rangle + \langle \phi | \hat{U}^\dagger(\theta) \hat{M} \frac{\partial \hat{U}(\theta)}{\partial \theta} | \phi \rangle \quad (84)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \frac{i}{2} \langle \phi | \hat{U}^\dagger(\theta) \hat{P} \hat{M} \hat{U}(\theta) | \phi \rangle - \frac{i}{2} \langle \phi | \hat{U}^\dagger(\theta) \hat{M} \hat{P} \hat{U}(\theta) | \phi \rangle \quad (85)$$

$$[\hat{P}, \hat{U}(\theta)] = 0 \implies \hat{P}\hat{U}(\theta) = \hat{U}(\theta)\hat{P} \quad (86)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \frac{i}{2} \langle \phi | \hat{U}^\dagger(\theta) [\hat{P}, \hat{M}] \hat{U}(\theta) | \phi \rangle \quad (87)$$

$$|\psi(\theta)\rangle = \hat{U}(\theta) |\phi\rangle \quad (88)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \frac{i}{2} \langle \psi(\theta) | [\hat{P}, \hat{M}] | \psi(\theta) \rangle \quad (89)$$

$$E(\theta + s) = \langle \psi(\theta) | e^{i\frac{s}{2}\hat{P}} \hat{M} e^{-i\frac{s}{2}\hat{P}} | \psi(\theta) \rangle \quad (90)$$

$$E(\theta + s) = \langle \psi(\theta) | \left[\cos^2\left(\frac{s}{2}\right) \hat{M} + i \sin\left(\frac{s}{2}\right) \cos\left(\frac{s}{2}\right) [\hat{P}, \hat{M}] + \sin^2\left(\frac{s}{2}\right) \hat{P} \hat{M} \hat{P} \right] | \psi(\theta) \rangle \quad (91)$$

$$E(\theta - s) = \langle \psi(\theta) | e^{-i\frac{s}{2}\hat{P}} \hat{M} e^{i\frac{s}{2}\hat{P}} | \psi(\theta) \rangle \quad (92)$$

$$E(\theta - s) = \langle \psi(\theta) | \left[\cos^2\left(\frac{s}{2}\right) \hat{M} - i \sin\left(\frac{s}{2}\right) \cos\left(\frac{s}{2}\right) [\hat{P}, \hat{M}] + \sin^2\left(\frac{s}{2}\right) \hat{P} \hat{M} \hat{P} \right] | \psi(\theta) \rangle \quad (93)$$

$$E(\theta + s) - E(\theta - s) = 2i \sin\left(\frac{s}{2}\right) \cos\left(\frac{s}{2}\right) \langle \psi(\theta) | [\hat{P}, \hat{M}] | \psi(\theta) \rangle \quad (94)$$

$$E(\theta + s) - E(\theta - s) = i \sin(s) \langle \psi(\theta) | [\hat{P}, \hat{M}] | \psi(\theta) \rangle \quad (95)$$

$$s = \frac{\pi}{2} \implies \sin\left(\frac{\pi}{2}\right) = 1 \quad (96)$$

$$E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) = i \langle \psi(\theta) | [\hat{P}, \hat{M}] | \psi(\theta) \rangle \quad (97)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \frac{1}{2} \left(i \langle \psi(\theta) | [\hat{P}, \hat{M}] | \psi(\theta) \rangle \right) \quad (98)$$

$$\frac{\partial E(\theta)}{\partial \theta} = \frac{1}{2} \left[E\left(\theta + \frac{\pi}{2}\right) - E\left(\theta - \frac{\pi}{2}\right) \right] \quad (99)$$

$$\theta_{n+1} = \theta_n - \eta \frac{\partial E(\theta_n)}{\partial \theta} \quad (100)$$

$$\theta^{(0)} = \frac{\pi}{4}, \quad s = \frac{\pi}{2} \quad (101)$$

$$\theta_+ = \theta^{(0)} + s = \frac{3\pi}{4} \quad (102)$$

$$P(1)_+ = \sin^2\left(\frac{3\pi}{8}\right) \approx 0.8535 \quad (103)$$

$$E(\theta_+) = 10(P(0)_+ - P(1)_+) = 10(0.1465 - 0.8535) = -7.07 \quad (104)$$

$$\theta_- = \theta^{(0)} - s = -\frac{\pi}{4} \quad (105)$$

$$P(1)_- = \sin^2\left(-\frac{\pi}{8}\right) \approx 0.1465 \quad (106)$$

$$E(\theta_-) = 10(P(0)_- - P(1)_-) = 10(0.8535 - 0.1465) = 7.07 \quad (107)$$

$$\frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{1}{2}(E(\theta_+) - E(\theta_-)) \quad (108)$$

$$\frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{1}{2}(-7.07 - 7.07) = -7.07 \quad (109)$$

$$\theta^{(1)} = \theta^{(0)} - \eta \frac{\partial E(\theta^{(0)})}{\partial \theta} = \frac{\pi}{4} + 7.07\eta \quad (110)$$

SPSA

$$X \sim \text{Bernoulli}(p = 0.5) \implies P(X = 1) = 0.5, \quad P(X = 0) = 0.5 \quad (111)$$

- X : variabel acak Bernoulli

$$\Delta_i = 2X - 1 \implies \Delta_i \in \{-1, 1\} \quad (112)$$

- Δ_i : variabel acak Rademacher komponen i

$$\vec{\Delta}_k = (\Delta_{k1}, \Delta_{k2}, \dots, \Delta_{kp})^T \quad (113)$$

- $\vec{\Delta}_k$: vektor perturbasi stokastik pada iterasi k

$$E_+ = E(\vec{\theta}_k + c_k \vec{\Delta}_k) \quad (114)$$

$$E_- = E(\vec{\theta}_k - c_k \vec{\Delta}_k) \quad (115)$$

- $\vec{\theta}_k$: vektor parameter pada iterasi k

- c_k : *perturbation step* pada iterasi k

$$E_+ = E(\vec{\theta}_k) + c_k \vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + \frac{1}{2} c_k^2 \vec{\Delta}_k^T \nabla^2 E(\vec{\theta}_k) \vec{\Delta}_k + O(c_k^3) \quad (116)$$

$$E_- = E(\vec{\theta}_k) - c_k \vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + \frac{1}{2} c_k^2 \vec{\Delta}_k^T \nabla^2 E(\vec{\theta}_k) \vec{\Delta}_k - O(c_k^3) \quad (117)$$

$$E_+ - E_- = 2c_k \vec{\Delta}_k^T \nabla E(\vec{\theta}_k) + O(c_k^3) \quad (118)$$

$$E_+ - E_- = 2c_k \sum_{j=1}^p \Delta_{kj} \frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^3) \quad (119)$$

$$\hat{g}_{ki} = \frac{E_+ - E_-}{2c_k \Delta_{ki}} \quad (120)$$

- $\hat{g}_{ki}$: estimator gradien komponen i pada iterasi k

$$\hat{g}_{ki} = \frac{2c_k \Delta_{ki} \frac{\partial E}{\partial \theta_i} + 2c_k \sum_{j \neq i}^p \Delta_{kj} \frac{\partial E}{\partial \theta_j} + O(c_k^3)}{2c_k \Delta_{ki}} \quad (121)$$

$$\hat{g}_{ki} = \frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + \sum_{j \neq i}^p \frac{\Delta_{kj}}{\Delta_{ki}} \frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^2) \quad (122)$$

$$\mathbb{E}[\hat{g}_{ki} | \vec{\theta}_k] = \mathbb{E} \left[\frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + \sum_{j \neq i}^p \frac{\Delta_{kj}}{\Delta_{ki}} \frac{\partial E(\vec{\theta}_k)}{\partial \theta_j} + O(c_k^2) \right] \quad (123)$$

$$\mathbb{E} \left[\frac{\Delta_{kj}}{\Delta_{ki}} \right] = \mathbb{E}[\Delta_{kj}] \mathbb{E} \left[\frac{1}{\Delta_{ki}} \right] \quad (124)$$

$$\mathbb{E}[\Delta_{kj}] = 0.5(1) + 0.5(-1) = 0 \quad (125)$$

$$\mathbb{E} \left[\frac{\Delta_{kj}}{\Delta_{ki}} \right] = 0 \cdot 0 = 0, \quad \forall j \neq i \quad (126)$$

$$\mathbb{E}[\hat{g}_{ki} | \vec{\theta}_k] = \frac{\partial E(\vec{\theta}_k)}{\partial \theta_i} + O(c_k^2) \quad (127)$$

$$\hat{H} = 10\hat{Z}_1 + 5\hat{Z}_2 \quad (128)$$

$$\hat{H} = 10(\hat{Z} \otimes \hat{I}) + 5(\hat{I} \otimes \hat{Z}) \quad (129)$$

Konfigurasi Awal:

- Hiperparameter: $a = 0.1, c = 0.1, \alpha = 0.602, \gamma = 0.101$

- Parameter Awal: $\vec{\theta}^{(0)} = [0.7854, 0.7854]^T$
- Fungsi Energi: $E(\vec{\theta}) = 10 \cos(\theta_1) + 5 \cos(\theta_2)$

Iterasi 1 ($k = 1$)

$$a_1 = \frac{0.1}{1^{0.602}} = 0.1, \quad c_1 = \frac{0.1}{1^{0.101}} = 0.1 \quad (130)$$

$$\vec{\Delta}_1 = [1, -1]^T \quad (131)$$

$$\vec{\theta}^+ = \vec{\theta}^{(0)} + c_1 \vec{\Delta}_1 = [0.8854, 0.6854]^T \quad (132)$$

$$\vec{\theta}^- = \vec{\theta}^{(0)} - c_1 \vec{\Delta}_1 = [0.6854, 0.8854]^T \quad (133)$$

$$E(\vec{\theta}^+) = 10 \cos(0.8854) + 5 \cos(0.6854) = 10.1995 \quad (134)$$

$$E(\vec{\theta}^-) = 10 \cos(0.6854) + 5 \cos(0.8854) = 10.9055 \quad (135)$$

$$\hat{g}_{11} = \frac{10.1995 - 10.9055}{2(0.1)(1)} = -3.53 \quad (136)$$

$$\hat{g}_{12} = \frac{10.1995 - 10.9055}{2(0.1)(-1)} = 3.53 \quad (137)$$

$$\vec{\theta}^{(1)} = \vec{\theta}^{(0)} - a_1 \hat{g}_1 = [1.1384, 0.4324]^T \quad (138)$$

$$E(\vec{\theta}^{(1)}) = 10 \cos(1.1384) + 5 \cos(0.4324) = 8.7306 \quad (139)$$

Iterasi 2 ($k = 2$)

$$a_2 = \frac{0.1}{2^{0.602}} = 0.0659, \quad c_2 = \frac{0.1}{2^{0.101}} = 0.0932 \quad (140)$$

$$\vec{\Delta}_2 = [-1, -1]^T \quad (141)$$

$$\vec{\theta}^+ = \vec{\theta}^{(1)} + c_2 \vec{\Delta}_2 = [1.0451, 0.3392]^T \quad (142)$$

$$\vec{\theta}^- = \vec{\theta}^{(1)} - c_2 \vec{\Delta}_2 = [1.2316, 0.5257]^T \quad (143)$$

$$E(\vec{\theta}^+) = 10 \cos(1.0451) + 5 \cos(0.3392) = 9.7331 \quad (144)$$

$$E(\vec{\theta}^-) = 10 \cos(1.2316) + 5 \cos(0.5257) = 7.6522 \quad (145)$$

$$\hat{g}_{21} = \frac{9.7331 - 7.6522}{2(0.0932)(-1)} = -11.1587 \quad (146)$$

$$\hat{g}_{22} = \frac{9.7331 - 7.6522}{2(0.0932)(-1)} = -11.1587 \quad (147)$$

$$\vec{\theta}^{(2)} = \vec{\theta}^{(1)} - a_2 \hat{g}_2 = [1.8735, 1.1676]^T \quad (148)$$

$$E(\vec{\theta}^{(2)}) = 10 \cos(1.8735) + 5 \cos(1.1676) = -1.0197 \quad (149)$$

Catatan: Karena $\Delta_{21} = \Delta_{22} = -1$, maka $\hat{g}_{21} = \hat{g}_{22}$. Estimator gradien SPSA tidak membedakan kontribusi parameter individual saat perturbasi seragam, berbeda dengan gradien analitik: $\nabla E = [-9.0795, -2.0954]^T$.

Ringkasan Konvergensi:

Iterasi (k)	θ_1	θ_2	$E(\vec{\theta})$	Jarak ke -15
0	0.7854	0.7854	10.6066	25.6066
1	1.1384	0.4324	8.7306	23.7306
2	1.8735	1.1676	-1.0197	13.9803